

Vinod A

Data Engineer

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Professional Summary

Proficient Azure Data Engineer with 4+ years of experience specializing in the healthcare and consumer goods company. Expertise in building scalable data pipelines and optimized data warehouses using Azure services and Databricks.

Technical Knowledge

- **Skills:** ETL, Data Warehousing, Data Profiling and validation, Azure, ETL Process optimization
- **Languages & Databases:** MS SQL Server, Oracle, SQL, Python, Pyspark
- **Tools:** Azure Synapse Analytics, Azure data factory, Databricks, Azure ADLS, Azure logic apps, Snowflake, Informatica PowerCenter, Putty
- **Libraries used:** logging, traceback, koalas, json, pandas, os, datetime, csv, delta table, sqlalchemy, concurrent.features, yaml, pyodbc. Pyspark.window, pyspark.functions, requests.
- **Support tools:** JIRA, Azure Devops

Professional Experience

Deloitte, Coimbatore

Consultant – Data Engineer | Apr 2024 - Present

- Worked as a Cloud Data Engineer for US based Healthcare Organization.
- Designed and developed an end-to-end, scalable **Master Data Ingestion Framework** within a single notebook, supporting ingestion from 100% of master data sources, reducing the number of **notebooks by 80%**, significantly cutting down manual intervention and maintenance effort.
- Implemented a unified, **dynamic merge logic** for dimension tables using an active indicator flag, enabling seamless handling of inserts and updates across **15+ dimension tables** and supporting onboarding of new sources with zero additional code changes.
- Built a robust, high-performance **Fact Data Ingestion pipeline** capable of processing millions of records daily, leveraging **checkpointing, broadcasting and other optimization** techniques to improve processing efficiency by up to 40% and reduce runtime variability.
- Developed a dependency-aware fact pipeline retriggering mechanism in Databricks, **automatically detecting upstream master data refreshes** and triggering downstream pipelines with zero manual effort.
- Implemented **comprehensive pipeline execution logging** by capturing detailed metadata including pipeline type, execution status, start/end timestamps, runtime metrics, and error messages.
- Integrated logging framework **with Azure SQL Database**, enabling real-time status updates to web application and robust auditing/monitoring capabilities.
- Used Git-based **version control (Azure Repos)** to manage ADF pipeline versions and collaborated with the team through pull requests and code reviews.

Cognizant Technology Solution, Coimbatore

Associate- Data Engineer | Jun 2021 – Mar 2024

- Developed a comprehensive end-to-end data ingestion pipeline with multi-configuration capabilities to ingest data from **10+ diverse sources including files, databases, and APIs** into a refined storage layer, processing over 5 TB of data monthly.
- Designed a reusable ingestion framework for handling **5+ file types with 100+ configurable functionalities (delimited, fixed-length, excel and parquet)**, reducing ingestion setup time by 40%.

- Built Azure Data Factory (ADF) and Synapse pipelines to support both incremental and full data loads from **various databases**, improving data load efficiency by 35%.
- Enabled client to reduce their infrastructure cost by 50% through the **implementation of parallel processing using python parallel processing and pyspark** which improved resource utilization and efficiency.
- Performed data refinement in PySpark, including **duplicate detection, null handling, and DDL validation, improving data quality and reducing downstream** errors by 50%.
- Implemented automated workflows and scheduling using ADF triggers with parameterization to support dynamic executions, increasing automation coverage by 60%.
- Logged pipeline execution metadata into SQL tables using ADF, enhancing traceability and reducing issue resolution time by 30%.
- Created automated test scenarios for ADF pipelines across data layers, reducing testing time by 70% and manual effort by 80%.

Education

Dr Mahalingam College of Engineering and Technology, Pollachi
Batchelor of Engineering — Electrical and Electronics
Anna University • Chennai • 2021 • 8.75

Certification

- Azure Fundamentals (AZ-900)
- Azure Data Fundamentals (DP-900)
- Azure Data Engineer Associate (DP -203)
- Databricks Data Engineer Associate